OCR GCSE Computer Science — **Ethical, Legal, Cultural and Environmental Impacts (J277/01 Section 1.6)**. This topic covers UK legislation affecting computer use (Data Protection Act, Computer Misuse Act, Copyright), ethical issues (privacy, AI, surveillance), cultural impacts (digital divide, automation), and environmental concerns (e-waste, energy consumption, sustainability).

Exam questions typically ask you to **describe** the purpose of specific laws, **explain** ethical issues around technology use, **discuss** environmental impacts of computing, and **apply** legislation to specific scenarios. The 2024 exam included questions on the Computer Misuse Act and environmental impacts of disposing of old hardware.

What you must know

This checklist covers the key exam-visible concepts. You should be able to explain each point and apply it to scenarios.

- **Describe** the purpose of the Data Protection Act 2018 and explain individuals' rights regarding their personal data.
- **Describe** the three offences under the Computer Misuse Act 1990.
- **Describe** how the Copyright, Designs and Patents Act 1988 protects software and digital content.

- **Explain** ethical issues related to computing: privacy, censorship, surveillance, AI decision-making.
- **Discuss** cultural impacts: digital divide, changes to employment, online communities.
- **Explain** environmental impacts: e-waste, energy consumption, carbon footprint, planned obsolescence.
- **Describe** strategies to reduce environmental impact: recycling, energy-efficient hardware, cloud computing.
- **Apply** legislation to scenarios and identify which law has been broken.

> **Specification note.** OCR J277 Section 1.6 requires understanding of legislation (Data Protection Act, Computer Misuse Act, Copyright), ethical issues, cultural impacts, and environmental concerns. Questions often present scenarios and ask you to identify which law applies or discuss pros and cons of technology use.

Toolkit

The three key UK laws

Law	Year	Purpose (one-line summary)	Key word
Data Protection Act	2018	Protects individuals' personal data and gives them rights over how it is used	Personal data
Computer Misuse Act	1990	Makes hacking, unauthorised access, and malware distribution illegal	Unauthorised access
Copyright, Designs and Patents Act	1988	Protects creators' rights over their original work (software, music, images)	Original work

Data Protection Act 2018 – key principles

The Data Protection Act 2018 implements GDPR in the UK. Organisations must follow these principles when handling personal data:

1. **Lawfulness, fairness, transparency** — data must be processed legally and openly
2. **Purpose limitation** — data collected for specific, stated purposes only
3. **Data minimisation** — only collect data that is necessary
4. **Accuracy** — data must be accurate and kept up to date
5. **Storage limitation** — data not kept longer than necessary
6. **Integrity and confidentiality** — data must be kept secure
7. **Accountability** — organisations must demonstrate compliance

Individual rights: Right to access your data, right to have errors corrected, right to have data deleted ("right to be forgotten"), right to know how data is being used.

Note: The Data (Use and Access) Act 2025 amends (but does not replace) the DPA 2018 and UK GDPR. The core principles above remain in force — DUAA mainly clarifies rules around research, automated decisions, and data sharing.

Computer Misuse Act 1990 – the three offences

This law was created to deal with hacking and computer-based crime. There are three main offences:

Offence	Description	Example	Maximum penalty
Section 1	Unauthorised access to computer material	Guessing someone's password to read their emails	2 years prison
Section 2	Unauthorised access with intent to commit further offences	Hacking into a bank to steal money	5 years prison
Section 3	Unauthorised modification of computer material	Spreading a virus, deleting files, installing malware	10 years prison

Key word: "Unauthorised" — if you have permission, it's not an offence under this act.

Copyright, Designs and Patents Act 1988

This law protects the creators of original work, giving them control over how their work is used, copied, and distributed.

What it protects:

- Software and computer programs
- Music, films, and video
- Images and photographs
- Written content (books, articles, websites)
- Databases

What is illegal:

- Copying software without permission (piracy)
- Downloading music/films illegally
- Using images without permission
- Distributing copyrighted material
- Removing copy protection

Licences: Software is typically sold with a licence that specifies how it can be used. Open-source software allows free use and modification; proprietary software restricts copying and modification.

Ethical vs Legal – understanding the difference

Legal = what the law says

- Defined by legislation
- Has specific penalties
- Enforced by courts
- Clear right/wrong

Ethical = what is morally right

- Based on values and principles
- No legal penalties
- Judged by society
- Often debatable

Key insight: Something can be legal but unethical (e.g., a company selling user data they collected with consent but in ways users didn't expect). Something can be illegal but arguably ethical (e.g., whistleblowing by leaking documents that expose wrongdoing).

Environmental impacts – key terms

Term	Definition	Computing example
E-waste	Electronic waste — discarded electrical devices	Old phones, computers, monitors dumped in landfill
Carbon footprint	Total greenhouse gas emissions caused by an activity	Energy used by data centres, manufacturing devices
Planned obsolescence	Designing products to become outdated or break after a certain time	Phones that slow down after updates, non-replaceable batteries
Rare earth metals	Scarce minerals needed for electronics	Lithium in batteries, cobalt, tantalum in capacitors

Term	Definition	Computing example
Sustainability	Meeting current needs without compromising future generations	Using renewable energy for data centres, recycling programs

1. Legislation – the laws you need to know

Laws exist to protect individuals, organisations, and society from harm caused by misuse of technology. You need to understand three key UK laws and be able to apply them to scenarios in the exam.

Data Protection Act 2018

Purpose: To protect individuals' personal data and give them control over how it is collected, stored, and used.

Personal data includes any information that can identify a living person: name, address, email, photos, IP addresses, health records, and more. The act requires organisations to:

- Only collect data for a **specific, stated purpose**
- Not keep data **longer than necessary**
- Keep data **accurate and up to date**
- Keep data **secure** from unauthorised access
- Not transfer data outside the UK without **adequate protection**

Individual rights include: accessing your data (Subject Access Request), correcting errors, deleting data ("right to be forgotten"), and objecting to automated decision-making.

Computer Misuse Act 1990

Purpose: To make hacking, unauthorised access, and spreading malware illegal.

Before this law, there was no specific offence for hacking into computer systems. The three offences are:

- **Section 1:** Unauthorised access — simply accessing a system without permission, even without causing damage (e.g., reading someone's emails)
- **Section 2:** Unauthorised access with intent to commit further offences — accessing a system to commit another crime (e.g., hacking to steal money or personal data)
- **Section 3:** Unauthorised modification — changing, deleting, or damaging data or software (e.g., spreading viruses, ransomware attacks, deleting files)

Key point: The word "unauthorised" is crucial. If an employee has permission to access a system, using it is not a Section 1 offence — but going beyond their authorised access could be.

Copyright, Designs and Patents Act 1988

Purpose: To protect creators' rights over their original work, including software,

music, images, and written content.

Copyright is automatic — as soon as you create an original work, you own the copyright. In computing, this protects:

- **Software** — the code and the program itself
- **Digital content** — music, videos, e-books, images
- **Databases** — collections of organised data
- **Website content** — text, graphics, layout

Piracy is the illegal copying or distribution of copyrighted material. This includes downloading films or music without paying, copying software without a licence, or sharing copyrighted files on peer-to-peer networks.

Exam tip — identifying which law applies.
Use these keywords to identify the correct law:

- **Personal data, privacy, consent, storing information** → Data Protection Act
- **Hacking, unauthorised access, virus, malware** → Computer Misuse Act
- **Copying, downloading, piracy, original work** → Copyright Act

If a scenario mentions someone accessing a system AND copying data, multiple laws may apply.

Exam tip — Section 1 vs Section 2 of Computer Misuse Act.
Section 1: Just accessing without permission (no further crime intended).
Section 2: Accessing with the **intent to commit another crime** (stealing money, data theft, blackmail).
The difference is **intent**. Reading someone's emails = Section 1. Reading their emails to find bank details to steal money = Section 2.

Freedom of Information Act 2000

While not always examined in detail, you should know that the Freedom of Information Act gives the public the right to request information held by public authorities (government, NHS, councils, police). This promotes transparency and accountability. Requests can be refused if disclosure would harm national security, commercial interests, or personal privacy.

Worked Example 1 – Identifying the correct law

Question: Tom downloads a film from an illegal website without paying. Which law has Tom broken? [1 mark]

Step 1: Identify the key action — downloading without paying = copying copyrighted material.
Step 2: Match to the law that protects original creative work.
Answer: Copyright, Designs and Patents Act 1988 [1 mark]

Exam tip: "Downloading without permission" or "copying" always points to Copyright. Don't confuse with Computer Misuse Act — that's about unauthorised access to systems, not downloading content.

Worked Example 2 – Computer Misuse Act scenario

Question: Sarah guesses her colleague's password and reads their private emails. State which section of the Computer Misuse Act Sarah has broken and explain why. [2 marks]

Step 1: Identify the action — accessing emails without permission.
Step 2: Determine intent — Sarah just read the emails, no further crime mentioned.
Answer: Section 1 — Unauthorised access to computer material [1 mark]. Sarah accessed the email system without permission/authorisation [1 mark].
Note: If Sarah had intended to use information from the emails to commit fraud, it would be Section 2.

Exam tip: Always state the section number AND explain why it applies. Just saying "Computer Misuse Act" without the section may not gain full marks.

Worked Example 3 – Data Protection Act application

Question: A company collects customer email addresses to send order confirmations but then sells this data to marketing companies without telling customers. Explain how this breaks the Data Protection Act. [3 marks]

Answer:

- The company collected data for a specific purpose (order confirmations) but used it for a different purpose (marketing) [1 mark]
- This breaks the principle of purpose limitation — data should only be used for the purpose it was collected [1 mark]
- The company did not have consent to share the data with third parties / lacked transparency about how data would be used [1 mark]

Exam tip: Reference specific principles (purpose limitation, consent, transparency) when explaining Data Protection Act breaches.

Worked Example 4 – Multiple laws scenario

Question: A hacker breaks into a hospital computer system, accesses patient medical records, and then sells this data online. Identify all the laws that have been broken and explain each. [4 marks]

Computer Misuse Act [2 marks]:

- Section 1: Unauthorised access to computer material (accessing the hospital system without permission) [1 mark]
- Section 2: Unauthorised access with intent to commit further offences (accessing to steal and sell data) [1 mark]

Data Protection Act [2 marks]:

- Medical records are sensitive personal data that must be kept secure [1 mark]
- Selling patient data without consent breaches data protection principles [1 mark]

Exam tip: Complex scenarios often break multiple laws. Check for: unauthorised access (Computer Misuse), personal data involved (Data Protection), and copying/distribution (Copyright).

2. Ethical and cultural impacts

Beyond the law, technology raises many ethical questions about what is right and wrong. These issues often have no clear answer — the exam may ask you to discuss both sides of an argument. Cultural impacts refer to how technology changes society, communities, and the way we live and work.

Privacy and surveillance

Key question: How much of our personal information should companies and governments be able to collect?

Arguments for surveillance/data collection:

- Helps prevent terrorism and serious crime
- Enables personalised services and recommendations
- Can improve public health (e.g., contact tracing)
- Helps find missing persons

Arguments against:

- Invasion of privacy — people have a right to private life
- Data can be misused or stolen
- Creates a "surveillance state" where everyone is monitored
- Chilling effect — people behave differently when watched

Artificial intelligence and automation

Key question: Should AI systems make important decisions about people's lives?

Ethical concerns about AI:

- **Bias** — AI can inherit biases from training data (e.g., facial recognition less accurate for some ethnicities)
- **Accountability** — who is responsible when AI makes a mistake?
- **Transparency** — AI decisions can be "black boxes" that can't be explained
- **Job displacement** — automation replacing human workers
- **Autonomous weapons** — should machines decide to use lethal force?

Benefits of AI:

- Can process data faster than humans
- Reduces human error in repetitive tasks
- Available 24/7 without fatigue
- Can detect patterns humans might miss

The digital divide

> **Definition:** The gap between those who have access to technology and those who do not.

The digital divide exists between:

- **Developed and developing countries** — access to internet infrastructure
- **Urban and rural areas** — broadband availability
- **Rich and poor** — ability to afford devices and connections
- **Young and old** — digital literacy and skills
- **Disabled and non-disabled** — accessibility of technology

Consequences: Those without access miss out on education, job opportunities, government services, social connections, and healthcare information.

Changes to employment

> **Key question:** How is technology changing the nature of work?

Jobs lost:

- Manufacturing automation (robots in factories)
- Self-service checkouts replacing cashiers
- AI replacing data entry and analysis roles
- Chatbots replacing customer service staff

Jobs created:

- Software developers and data scientists
- Cybersecurity specialists
- Social media managers
- AI trainers and ethicists
- Remote work opportunities

Changed work patterns: Working from home, gig economy, 24/7 connectivity blurring work-life boundaries.

Censorship and freedom of expression

The internet has enabled unprecedented freedom to share information and opinions. However, this raises questions about what content should be allowed online.

Arguments for some censorship:

- Protecting children from harmful content
- Preventing spread of misinformation
- Stopping hate speech and incitement to violence
- Protecting national security
- Removing illegal content (terrorism, child abuse)

Arguments against censorship:

- Freedom of speech is a fundamental right
- Governments could abuse censorship powers
- Difficult to define what is "harmful"
- Censorship can suppress legitimate political opposition
- Information often finds other ways to spread

Worked Example 5 — Discussing privacy vs security

Question: A government proposes that all messaging apps should allow police to read encrypted messages to help catch criminals. Discuss the ethical implications of this proposal. [6 marks]

Arguments in favour [3 marks]:

- Could help police investigate serious crimes like terrorism or child exploitation [1 mark]
- Criminals currently use encryption to hide illegal activities [1 mark]
- May help protect public safety and prevent attacks [1 mark]

Arguments against [3 marks]:

- Breaks encryption for everyone, not just criminals, reducing privacy for all users [1 mark]
- Could be abused by governments to spy on citizens or political opponents [1 mark]
- Creates security vulnerabilities that hackers could exploit [1 mark]

Exam tip: "Discuss" questions require BOTH sides. Aim for 3 points on each side for a 6-mark question. Don't just give your opinion — present balanced arguments.

Worked Example 6 — AI bias scenario

Question: A bank uses an AI system to decide whether to approve loan applications. Explain two ethical concerns about using AI for this purpose. [4 marks]

Concern 1 [2 marks]:

The AI may contain bias from its training data [1 mark]. If historical data shows certain groups were less likely to be approved (even unfairly), the AI will continue this discrimination [1 mark].

Concern 2 [2 marks]:

AI decisions can be difficult to explain or appeal [1 mark]. Applicants may not understand why they were rejected and have no way to challenge an unfair decision / lack of transparency [1 mark].

Alternative concerns:

- Accountability — unclear who is responsible if AI makes wrong decisions
- Job losses — replacing human loan officers
- Dehumanising — important decisions made without human consideration

Exam tip: Always give the concern AND explain the consequence. "AI might be biased" alone is only 1 mark. Explain what happens as a result.

Worked Example 7 – Digital divide explanation

Question: Explain what is meant by the "digital divide" and describe two consequences for people who lack access to technology. [4 marks]

Definition [1 mark]:
The digital divide is the gap between people who have access to computers/internet and those who do not [1 mark].

Consequence 1 [1-2 marks]:
Limited access to education — many learning resources, courses, and homework submissions are online [1 mark], so students without access fall behind / cannot complete work [1 mark].

Consequence 2 [1-2 marks]:
Reduced job opportunities — many jobs require digital skills or online applications [1 mark], so people without access are excluded from the job market [1 mark].

Exam tip: Other valid consequences include: missing government services (benefits, tax), social isolation, missing health information, unable to shop online (often cheaper).

Worked Example 8 – Automation and employment

Question: A supermarket is replacing checkout staff with self-service machines. Discuss the impacts of this change. [4 marks]

Negative impacts [2 marks]:

- Job losses for checkout staff / unemployment [1 mark]
- Difficult for elderly or less tech-confident customers [1 mark]
- Loss of human interaction / less personal service
- May increase shoplifting as fewer staff monitoring

Positive impacts [2 marks]:

- Faster checkout for customers with few items [1 mark]
- Reduced costs for the business / savings can lower prices [1 mark]
- Staff can be redeployed to other roles (stacking, customer help)
- Available 24/7 without needing night staff

Exam tip: "Discuss the impacts" requires both positive and negative points. Even if you personally disagree with automation, present both sides.

3. Environmental impacts

Computing has significant environmental impacts, from the energy consumed by data centres to the mountains of electronic waste created by discarded devices. Understanding these impacts and potential solutions is increasingly important — and increasingly examined.

E-waste – the problem

E-waste (electronic waste) refers to discarded electrical and electronic devices.

Why e-waste is a problem:

- **Toxic materials** — devices contain lead, mercury, cadmium, and other hazardous substances that can leak into soil and water
- **Volume** — over 50 million tonnes of e-waste generated globally each year
- **Improper disposal** — much e-waste is shipped to developing countries where it is processed unsafely
- **Resource waste** — valuable materials (gold, copper, rare earth metals) are lost when devices are not recycled properly

What contributes to e-waste:

- Frequent upgrades (new phone every 1-2 years)
- Planned obsolescence
- Devices designed to be difficult to repair
- Non-replaceable batteries
- Software updates that slow old devices

Energy consumption and carbon footprint

Carbon footprint is the total greenhouse gas emissions caused by an activity or product.

Where computing uses energy:

- **Data centres** — servers running 24/7 consume massive amounts of electricity for processing AND cooling
- **Manufacturing** — producing devices requires energy and raw materials
- **Networks** — transmitting data across the internet uses energy
- **End-user devices** — phones, laptops, PCs charging and running
- **Cryptocurrency mining** — extremely energy-intensive calculations

Scale of the problem: Data centres alone account for approximately 1-2% of global electricity use. A single Google search uses enough energy to power a 60W light bulb for 17 seconds. Streaming video is particularly energy-intensive.

Reducing environmental impact

Strategies for individuals:

- **Keep devices longer** — resist upgrading when current device still works
- **Recycle properly** — use designated e-waste collection points
- **Repair instead of replace** — fix broken devices where possible
- **Buy refurbished** — extends device lifespan
- **Reduce energy use** — turn off devices, use power-saving modes
- **Stream less in HD** — lower video quality uses less energy

Strategies for organisations:

- **Renewable energy** — power data centres with solar, wind
- **Efficient cooling** — use natural cooling, locate in cold climates
- **Cloud computing** — share resources efficiently rather than each company running idle servers
- **Modular design** — devices that can be upgraded rather than replaced
- **Take-back schemes** — manufacturers accepting old devices for recycling

Positive environmental uses of technology

Technology can also **help** the environment:

- **Smart grids** — optimise electricity distribution, reduce waste
- **Remote working** — reduces commuting and office energy use
- **Video conferencing** — reduces need for travel
- **Environmental monitoring** — sensors tracking pollution, deforestation
- **Precision agriculture** — reduces water and pesticide use
- **Paperless offices** — reduces paper consumption
- **Climate modelling** — simulations helping understand climate change
- **Smart heating/lighting** — reduces energy waste in buildings

> **Exam tip — balanced answers.**
> If asked about environmental impacts, remember technology can be both harmful AND helpful. A good answer acknowledges both sides.

Planned obsolescence

Planned obsolescence is when products are deliberately designed to have a limited lifespan, either through physical degradation or becoming outdated.

Types of planned obsolescence:

- **Technical** — components designed to fail after a certain time
- **Software** — updates that slow down old devices or remove support
- **Style** — new designs making old products seem outdated

- **Functional** — new features that make old versions seem inadequate

Why companies do it:

- Encourages customers to buy new products more frequently
- Increases revenue and profits
- Allows faster innovation cycles

Ethical concerns:

- Creates unnecessary waste
- Wastes resources and energy
- Costs consumers more money
- Often hidden from consumers (deceptive)

Worked Example 9 — E-waste disposal [2024 exam style]

Question: A school is replacing all its old computers with new ones. Describe two environmental impacts of disposing of the old computers and explain how these impacts could be reduced. [6 marks]

Impact 1 + reduction [3 marks]:

- Old computers contain toxic materials (lead, mercury) that can leak into soil and water if sent to landfill [1 mark]
- This could harm wildlife and contaminate drinking water [1 mark]
- Reduction: Use a certified e-waste recycling company that safely extracts and disposes of hazardous materials [1 mark]

Impact 2 + reduction [3 marks]:

- Valuable materials (copper, gold, rare earth metals) are wasted if computers are not recycled [1 mark]
- More mining is needed to extract new raw materials, damaging environments [1 mark]
- Reduction: Donate working computers to charities or schools that need them / sell for refurbishment [1 mark]

Exam tip: This type of question follows a pattern: state the impact, explain the consequence, then give a practical solution. Practice this structure.

Worked Example 10 — Data centre energy consumption

Question: Large technology companies operate data centres that consume significant amounts of energy. Explain two ways these companies can reduce the environmental impact of their data centres. [4 marks]

Method 1 [2 marks]:

Use renewable energy sources such as solar or wind power to run the data centre [1 mark]. This reduces carbon emissions as no fossil fuels are burned to generate electricity [1 mark].

Method 2 [2 marks]:
Locate data centres in cold climates or use efficient cooling systems [1 mark]. This reduces the energy needed for air conditioning / cooling accounts for a large proportion of data centre energy use [1 mark].

Alternative methods:

- Virtualisation — running multiple virtual servers on fewer physical machines
- Using waste heat — redirecting heat to warm nearby buildings
- Efficient hardware — using processors designed for energy efficiency

Exam tip: Always explain WHY the method reduces environmental impact. "Use renewable energy" alone is 1 mark. Adding "reduces carbon emissions from burning fossil fuels" gains the second mark.

Worked Example 11 – Planned obsolescence ethics

Question: Some technology companies are accused of "planned obsolescence" — designing products to become outdated or stop working after a certain time. Discuss the ethical implications of this practice. [6 marks]

Arguments that it is unethical [3 marks]:

- Deliberately shortening product life creates unnecessary e-waste that harms the environment [1 mark]
- Consumers are deceived into thinking they need to upgrade when their device could last longer [1 mark]
- Forces consumers to spend more money than necessary / exploitative [1 mark]

Arguments in defence / counterpoints [3 marks]:

- Frequent replacements drive innovation and bring new features to consumers faster [1 mark]
- Companies need to make profits to continue developing products and employing workers [1 mark]
- Sometimes ending support for old devices is necessary for security reasons [1 mark]

Exam tip: Ethics questions rarely have a "right" answer. Present both sides fairly, even if you have a strong personal opinion.

Worked Example 12 – Technology helping the environment

Question: While technology can harm the environment, it can also help protect it. Describe two ways technology is being used to reduce environmental damage. [4 marks]

Way 1 [2 marks]:
Smart thermostats and lighting systems can automatically adjust heating and lighting based on occupancy [1 mark]. This reduces energy waste by not heating or lighting empty rooms [1 mark].
Way 2 [2 marks]:
Video conferencing allows people to have meetings without travelling [1 mark]. This reduces carbon emissions from cars and planes / reduces fuel consumption [1 mark].
Alternative examples:

- Environmental monitoring sensors tracking pollution, deforestation
- Electric vehicles and smart charging infrastructure
- Precision agriculture reducing water and pesticide use
- Digital documents reducing paper consumption

Exam tip: Give specific, concrete examples rather than vague statements. "Technology can monitor things" is weak. "Sensors can track air pollution levels to identify problem areas" is stronger.

Summary of worked examples

#	Topic	Question type	Key skill
1	Copyright Act	Identify law	Match scenario to legislation
2	Computer Misuse Act	Section identification	Distinguish Section 1 vs 2 vs 3
3	Data Protection Act	Explain breach	Reference specific principles
4	Multiple laws	Complex scenario	Identify all applicable laws
5	Privacy vs security	Discussion	Present both sides of argument
6	AI ethics	Explain concerns	Bias, accountability, transparency
7	Digital divide	Define + consequences	Access gap + real impacts
8	Automation	Discuss impacts	Positive and negative effects
9	E-waste (2024 style)	Impact + solution	Toxic materials, proper disposal
10	Data centres	Reduction strategies	Renewable energy, cooling
11	Planned obsolescence	Ethics discussion	Both sides of debate

#	Topic	Question type	Key skill
12	Tech helping environment	Describe benefits	Specific, concrete examples

Quick reference

Key terms

Term	Definition (exam-answer style)
Data Protection Act	UK law that protects individuals' personal data and gives them rights over how it is collected, stored, and used
Computer Misuse Act	UK law that makes unauthorised access to computer systems, and modification of data, illegal
Copyright	Legal protection for creators of original work, preventing others from copying without permission
Personal data	Any information that can identify a living individual (name, address, email, photos, IP address)
Piracy	Illegally copying or distributing copyrighted material
Privacy	The right to keep personal information and activities private from others
Digital divide	The gap between those with access to technology and those without
E-waste	Discarded electronic devices and equipment
Carbon footprint	Total greenhouse gas emissions caused by an activity or product
Planned obsolescence	Deliberately designing products to become outdated or stop working after a certain time
Sustainability	Meeting current needs without compromising the ability of future generations to meet their needs
Ethical	Relating to moral principles of right and wrong behaviour

Laws at a glance

Law	Protects against	Key scenario triggers
Data Protection Act 2018	Misuse of personal data	Collecting, storing, sharing personal information

Law	Protects against	Key scenario triggers
Computer Misuse Act 1990	Hacking and malware	Unauthorised access, viruses, modifying data
Copyright Act 1988	Theft of creative work	Copying software, downloading music/films illegally

Computer Misuse Act sections

Section	Offence	Memory aid
Section 1	Unauthorised access	Just looking (no further crime)
Section 2	Access with intent to commit further offence	Looking to steal/commit crime
Section 3	Unauthorised modification	Changing/deleting/damaging (viruses, ransomware)

Environmental impacts summary

Problem	Cause	Solution
E-waste	Frequent device upgrades, planned obsolescence	Recycle properly, repair, buy refurbished
Energy consumption	Data centres, device charging, streaming	Renewable energy, efficient cooling, power-saving modes
Resource depletion	Mining rare earth metals for components	Recycling to recover materials, modular design
Carbon emissions	Manufacturing, shipping, powering devices	Longer device lifespans, remote working

Typical pitfalls

Pitfall 1 — Confusing laws.
Wrong: "Downloading a pirated film breaks the Computer Misuse Act."
Right: Piracy breaks the **Copyright, Designs and Patents Act**. Computer Misuse Act is about **unauthorised access**, not copying content.

Pitfall 2 — Not specifying CMA section.
Wrong: "This breaks the Computer Misuse Act."

Right: State which **section**: "This breaks **Section 1** of the Computer Misuse Act — unauthorised access to computer material."

Pitfall 3 — Mixing up Section 1 and Section 2.
Wrong: "Reading someone's emails without permission is Section 2."
Right: Just accessing = **Section 1**. Accessing **with intent to commit another crime** = Section 2.

Pitfall 4 — Confusing ethical and legal.
Wrong: "It's unethical because it's against the law."
Right: Ethical = about morals/values. Legal = about laws. Something can be legal but unethical, or illegal but arguably ethical.

Pitfall 5 — One-sided discussion answers.
Wrong: Only giving arguments for OR against a topic.
Right: "Discuss" questions need **BOTH sides**. Even if you disagree with one side, present it fairly.

Pitfall 6 — Vague environmental answers.
Wrong: "Technology is bad for the environment."
Right: Be specific: "Manufacturing devices requires mining **rare earth metals**, which damages habitats and uses large amounts of **energy**."

Pitfall 7 — Forgetting the "why" in solutions.
Wrong: "Companies should use renewable energy."
Right: "Companies should use renewable energy **because this reduces carbon emissions from burning fossil fuels**."

Pitfall 8 — Thinking digital divide is only about money.
Wrong: "The digital divide is about rich and poor."
Right: Also includes: urban/rural, young/old, different countries, disability accessibility, digital skills.

Pitfall 9 — Not linking AI concerns to consequences.
Wrong: "AI might be biased."
Right: "AI might be biased, **which could lead to unfair decisions affecting people's loans, jobs, or legal outcomes**."

Pitfall 10 — Forgetting technology can help the environment.
Wrong: Only discussing negative environmental impacts.

Right: Technology also **helps**: video conferencing reduces travel, smart systems reduce energy waste, sensors monitor pollution.

Assessment Hub | Long-Answer | Notes Index

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.